

Tasks

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.



Steps Performed

1. Opened Wireshark and started capture on the active Wi-Fi interface.
2. Loaded www.spotify.com in the browser.
3. Stopped capture after the page finished loading.
4. Applied filters to identify DNS and HTTP/HTTPS packets.

Observed Packets

- DNS Packets
 - Query: Standard query A www.spotify.com
 - Response: Standard query response A 35.186.xxx.xxx
- HTTPS Packets
 - Client Hello: TLSv1.2 Client Hello to www.spotify.com
 - GET Request: GET / HTTP/1.1 Host: www.spotify.com

2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.



Step 1: Command Used

Code

```
nslookup www.flipkart.com
```

Step 2: Output (Key Part)

- Server: [Your DNS server, e.g., 8.8.8.8]
- Address: [DNS server IP]
- Non-authoritative answer:
 - Name: www.flipkart.com
 - Address: 163.53.76.86

Explanation

- DNS Query: Your computer asked the DNS server to resolve the domain www.flipkart.com.
- DNS Response: The server replied with the IPv4 address 163.53.76.86.
- Meaning: This IP is the actual location of Flipkart's web server on the internet. When you visit Flipkart, your browser uses this IP to establish a connection.

3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.



Aspect	HTTP	HTTPS
Security	No encryption; data sent in plain text	Encrypted using SSL/TLS, protects against eavesdropping
Port Number	Default port 80	Default port 443
Trust & Authentication	No certificate required	Requires a valid SSL/TLS certificate to verify website identity

Real-World Example

- Instagram app / website → relies on HTTPS to ensure your login credentials, messages, and photos are transmitted securely.
- Without HTTPS, attackers could intercept sensitive data over public Wi-Fi.

4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.
Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.



- Opened Wireshark and applied the filter:

Code

dns

- In the browser, visited a new website (not previously cached).
- Observed the DNS traffic sequence in Wireshark.

Sequence Observed

- DNS Query (Standard query A) → Your computer asks the DNS server for the IP address of the new website.
- DNS Response (Standard query response A) → DNS server replies with the IPv4 address.
- DNS Query (Standard query AAAA) → Your computer may also request the IPv6 address.
- DNS Response (Standard query response AAAA) → DNS server replies with the IPv6 address if available.

5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).



Protocol Chosen: DHCP (Dynamic Host Configuration Protocol)

Scenario: Imagine you walk into a café, open your phone, and connect to the café's Wi-Fi.

- As soon as you join, your phone doesn't know what IP address to use.
- The DHCP server in the café's router automatically assigns your phone:
 - An IP address (so it can be identified on the network)
 - A default gateway (so traffic can reach the internet)
 - A DNS server (so you can resolve names like *instagram.com*)

Now, when you open Instagram, your phone already has all the network settings it needs — thanks to DHCP — and you can scroll through posts without manually configuring anything.